



Introduction to Microeconomics Fall 2025

SEMINAR #5

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Exercise 1

Each firm in a perfectly competitive market has this total cost function: $TC(Q) = 16 + Q^2$, and the market demand is given by: $Q^d(P) = 24 - P$.

In the long-run equilibrium, calculate:

1. The price and quantity produced by each firm
2. The total quantity produced
3. The number of firms in the market in the long-run equilibrium.

Now imagine that the market were a monopoly and not perfectly competitive. The monopolist has the total cost function given above, calculate:

1. Marginal revenue
2. Quantity produced
3. Price

Graphically represent the situation, calculate:

4. The monopolist's profit
5. The net welfare loss relative to the situation where the monopolist is forced to charge marginal cost

7) PERFECT COMPETITION
 $MR = MC$

$$\begin{aligned} TC(q) &= 16 + q^2 \\ MC(q) &= 2q \end{aligned}$$

DEMAND:
 $Q = 24 - P$
 $\rightarrow P = 24 - Q$

$$MR = P = MC = 2q$$

$$TR = TC$$

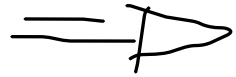
PLUG IN FOR PRICE

$$PQ = 16 + q^2$$

$$\rightarrow 2q \cdot q = 16 + q^2 \rightarrow 2q^2 = 16 + q^2 \rightarrow q^2 = 16 \rightarrow q^* = 4$$

$$q^* = 4 \xrightarrow{\text{PLUG IN MARGINAL COST TO FIND EQ. PRICE}} P$$

$$P^* = MC = 2q^* = 2 \cdot 4 = 8$$



PLUG EQ. PRICE INTO MARKET DEMAND TO FIND MARKET SUPPLY

$$Q^* = 24 - P^* = 24 - 8 = 16$$



FIND TOTAL NUMBER OF FIRMS

$$Q = n \cdot q \rightarrow n = \frac{Q^*}{q^*} = \frac{16}{4} = 4$$

2) MONOPOLY

$$2) TR = PQ = (24 - Q) \cdot Q = 24Q - Q^2$$

$$MR = 24 - 2Q \quad P = 24 - Q = 24 - 6 = 18$$

$$MR = MC$$

$$4Q = 24$$

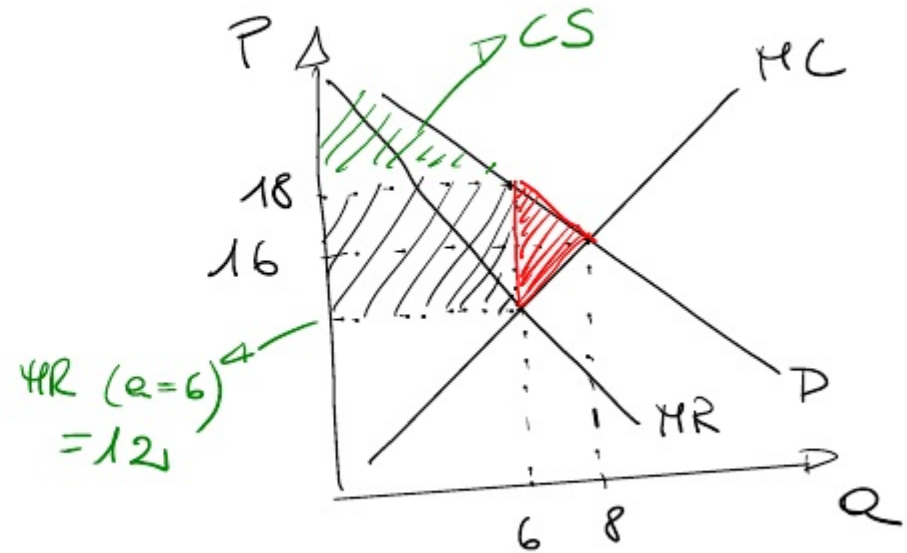
$$Q^* = 6$$

$$\pi = TR - TC = PQ - 16 - Q^2$$

$$= 18 \cdot 6 - 16 - 36 = 108 - 62 = 46$$

$$LOSS = \frac{(18 - 12) \cdot (8 - 6)}{2} = 6$$

$$MR = MC = 2Q$$



Exercise 2

$$\begin{aligned} Q &= 10 - P \rightarrow P = 10 - Q \\ TC &= 10 \\ MR &= MC \\ TC &= 10 \rightarrow MC = 0 \end{aligned}$$

$$\begin{aligned} TR &= PQ = Q(10 - Q) = 10Q - Q^2 \\ MR &= 10 - 2Q = 0 \\ Q^* &= \frac{10}{2} = 5 \end{aligned}$$

Elon AG proposes to supply electricity to an island with revolutionary wind turbines that require no maintenance. The total cost of his company is given by $TC = 10$. The demand for electricity on this island is given by the function $Q^d = 10 - P$. The managers give him the exclusive right to supply electricity, which makes it a monopoly. At equilibrium, which statement is correct?

a) The total expenditure on electricity on the island is 25. ✓

b) The marginal cost is less than the marginal revenue. ✗

c) The firm Elon AG makes no profit because the equilibrium price is 0. ✗

d) The firm Elon AG makes no profit because the quantity produced at equilibrium is 0. ✗

$$P^* = 10 - 5 = 5$$

$$TE = P^* \cdot Q^* = 25$$

$$P = 6 - Q$$

$$TR = Q(6 - Q) = 6Q - Q^2$$

$$MR = 6 - 2Q$$

$$MC = Q$$

$$MR = MC$$

$$6 - 2Q = Q \quad 6 = 3Q$$

$$Q^* = 2$$

$$P^* = 6 - 2 = 4$$

Exercise 3

Consider a market where demand is represented by the function $Q^d = 6 - P$. A firm, in a monopoly situation, faces the following cost function: $TC = \frac{Q^2}{2} + 5$. The market is in equilibrium but a legislator proposes to force the firm to apply marginal cost pricing. By how much would the producer surplus vary?

a) +0.5.

b) -1.

✓ c) -1.5.

d) -2.

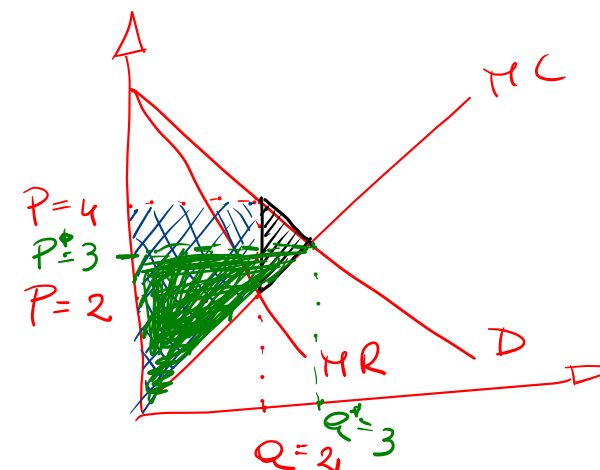
$$PS^{\text{monopoly}} = \frac{(4-2) \cdot 2 + (2 \cdot 2)}{2} = 4 + 2 = 6$$

$$Q = 6 - Q \quad 2Q = 6 \quad Q^* = 3$$

$$P^* = 3$$

$$PS^{\text{comp}} = \frac{P \cdot Q}{2} = \frac{9}{2} = 4.5$$

$$\Delta PS = 6 - 4.5 = -1.5$$



Exercise 4

A monopolist has marginal revenue and marginal cost of \$19 at 15 units of output. At this quantity, the price on the demand curve is \$38 and the average total cost is \$15. What is the firm's profit?

a) \$255

b) \$345 ✓

c) \$19

d) \$285

$$MR = MC = 19$$

$$Q = 15$$

$$P = 38$$

$$ATC = 15$$

$$\pi = TR - TC$$

$$= P \cdot Q - Q \cdot ATC$$

$$= Q (P - ATC)$$

$$= 15 (38 - 15)$$

$$= 15 \cdot 23$$

$$= 345$$

$$\text{Lerner Index} = \frac{P - MC}{P} = \frac{1}{|\epsilon_p^D|}$$

Exercise 5

A profit-maximizing monopoly faces decreasing demand that has an equilibrium price elasticity of -3 . The firm's marginal cost at the optimal level of output is 10. What price is optimal for the monopoly?

a) 25.

b) 15. ✓

c) 6.

d) 5.

$$\hookrightarrow \frac{P - 10}{P} = \frac{1}{3}$$

$$3(P - 10) = P$$

$$2P = 30$$

$$P^* = 15$$

$$MC = 3 \rightarrow MR$$

Exercise 6

A movie theater has 1000 potential customers, 900 students, and 100 professors. The willingness to pay for an admission ticket for a student is 5 CHF and 10 CHF for a professor. Each person can buy only one ticket. The marginal cost of a ticket is constant and equal to 3 CHF and there is no fixed cost. Which of the following statements is correct?

- ✓ a) The firm's profit will be maximized if it can charge different prices for each group of consumers and charge 5 CHF to a student and 10 CHF to a professor.
- ✗ b) The firm's profit will be maximized if it charges a uniform price of 10 CHF. $\rightarrow \pi(p=10) = 10 \cdot 100 = 1000 - 300$
- ✗ c) The firm's profit will be maximized if it charges a uniform price of 3 CHF. $\rightarrow \pi(p=5) = 5 \cdot 1000 = 5000 - 3000$
- ✗ d) A price of 5 CHF for a student and 10 CHF for a professor is an example of second-degree price discrimination.

Exercise 7

Consider a monopolistic firm with the following total cost function:

$$TC(Q) = 9/2 Q^2 + 12 \quad \rightarrow MC = 9Q$$

The firm faces a demand represented by the function:

$$Q^D(P) = -2P + 80 \quad \rightarrow P = \frac{80 - Q}{2} \quad P^* = \frac{76}{2} = 38$$

Which of the following statements is correct?

- ✓ a) The equilibrium profit is 68.
- ✗ b) The equilibrium quantity is 4 and the price is 36.
- ✗ c) The marginal cost is 9Q and the marginal revenue is 40 - Q/2.
- ✗ d) The equilibrium profit is 60.

$$\begin{aligned} TR &= Q \cdot P \\ &= Q \left(\frac{80 - Q}{2} \right) \quad \rightarrow MR = 40 - Q \\ &= 40Q - \frac{Q^2}{2} \end{aligned}$$

$$9Q = 40 - Q$$

$$10Q = 40$$

$$Q^* = 4$$

$$\begin{aligned} \pi &= TR - TC = 38 \cdot 4 - \left(\frac{9}{2} 4^2 + 12 \right) \\ &= 68 \end{aligned}$$

$$\begin{aligned}\epsilon^A &= -2 \\ \epsilon^B &= -4\end{aligned}$$

$$\frac{P - MC}{P} = \frac{1}{|\epsilon|}$$

Exercise 8

Consider a firm that sells the same product in two disjoint market segments. The marginal cost of production is the same in both markets. In segment A, the price elasticity of demand for its product is equal to -2. In segment B, this elasticity is -4. What will be the pricing strategy that will maximize its profits?

- ✗ a) Apply the same price (monopoly price) on both segments.
- ✓ b) Apply a higher price on segment A and a lower price on segment B.
- ✗ c) Apply a higher price on segment B and a lower price on segment A.
- ✗ d) Put a price lower than the marginal cost on both segments.

$$\frac{P_A - MC}{P_A} = \frac{1}{2} \Rightarrow P_A = 2MC$$

$$\frac{P_B - MC}{P_B} = \frac{1}{4} \Rightarrow P_B = \frac{4}{3} MC$$

$$P_A > P_B$$

Exercise 9

In an oligopolistic market, which of the following statements is correct?

- ☒ a) Firms are in a "strategic game" situation where the Nash equilibrium is Pareto optimal.
- ☒ b) It is not optimal for firms to cooperate.
- ☒ c) Creating a cartel is optimal for firms.
- ☒ d) Only strategies with punishment can maintain the Nash equilibrium in this type of market.

Exercise 10

In the presence of a cartel:

- ✓ a) The total quantity exchanged on the market corresponds to that resulting from a monopoly situation and remains lower than that obtained in competition.
- ✗ b) Each firm produces a quantity equal to that which it would produce in a monopoly situation.
- ✗ c) Consumer surplus is greater than in a monopoly situation.
- ✗ d) Each firm tends to reduce its individual supply in order to increase the price and its total revenue.

Exercise 11

The University of Montana and Wyoming are competing to attract prospective students from the American Northwest. Each institution is simultaneously choosing to raise or lower its tuition fees. Depending on the strategy implemented and that of the competing university, tuition revenue varies considerably. The following matrix presents the different scenarios for tuition revenue (in thousands of US dollars):

At the Nash equilibrium, what strategy will the University of Montana and Wyoming play?

- a) Increase; Increase
- b) Decrease; Increase
- c) Increase; Decrease
- d) Decrease; Decrease

Montana

~~RAISE~~
Augmenter
~~LOWER~~
Baisser

Wyoming

~~RAISE~~
Augmenter
~~LOWER~~
Baisser

	Augmenter	Baisser
Augmenter	(100, 40)	(60, 80)
Baisser	(160, 80)	(20, 120)

Exercise 12

Rokhaya and Myriam want to spend their Sunday together but they have different preferences regarding the choice of activity. Rokhaya wants to go to the countryside while Myriam prefers to go to the beach. The table below summarizes their level of well-being for each situation

Determine the Nash equilibrium(s) of this game.

- ✓ a) (Countryside; Countryside) and (Beach; Beach).
- b) (Beach; Countryside) and (Countryside; Beach).
- c) (Countryside; Countryside).
- d) There is no equilibrium in this game.

Rokhaya

Aller à la campagne

Aller à la plage

Myriam

Aller à la campagne

Aller à la plage

	Aller à la campagne	Aller à la plage
Aller à la campagne	(3,2)	(1,1)
Aller à la plage	(1,1)	(2,3)